

Anand Shankar

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SUMMARY

Software Engineer with **5+ years** of experience building scalable **cloud services**, high-throughput analysis pipelines and **full-stack applications** across research and industry environments. Shipped **production-grade software** across the **full SDLC** at Illumina and NIH - from **system architecture** and **CI/CD automation** to **deployment on AWS**. Hands-on experience in **Python, C++ and Java** with a strong foundation in **object-oriented design**, large-scale data processing and **cross-functional collaboration** with research and engineering teams.

SKILLS

Programming: Python (Pytorch, Tensorflow, Pandas), Java, C++, SQL, Javascript, R, Linux/Bash

Frameworks & Tools: Django, React, Node.js, GraphQL, Jenkins, Docker, Git, Snakemake, Nextflow

Cloud Platforms: AWS (S3, EC2, Batch), Azure

WORK EXPERIENCE

National Institutes of Health (National Cancer Institute)

Aug 2023 – Present | Remote

Leidos Biomedical Research, Frederick National Laboratory

Analyst II

- Led **benchmarking** and **performance evaluation** of FPGA/GPU-accelerated compute pipelines (Illumina DRAGEN, NVIDIA Parabricks) against internal baselines, demonstrating **8-10x runtime improvements** and **2-3x reduction in error rates**, leading to informed infrastructure investment decisions.
- Built and maintained **scalable software pipelines** using **Snakemake** and **Nextflow** on **HPC** and **AWS**, processing **>1TB** of high throughput DNA sequencing data, delivering reliable high-throughput data processing capabilities for NCI research initiatives.
- Containerized and deployed cloud services on **AWS** using **Docker**, standardizing pipeline execution environments across NCI/NIH compute infrastructure and improving reliability, scalability, and reproducibility of production workflows.

Illumina

Mar 2022 – Aug 2023 | San Diego, CA

Software Engineer II

- Designed, implemented and deployed production-grade **Nextflow pipelines** integrating DRAGEN multi-omic workflows with containerized **Python/R modules**, ensuring workflow robustness and contributing to a **25% increase** in ICA (Illumina Connected Analytics) platform adoption.
- Developed **CI/CD pipelines** with **Jenkins**, automating deployment and **regression testing**, **reducing pipeline release times by 40%** and improving software quality.
- Resolved high-priority customer issues by conducting simulations on **HPC** and **cloud platforms (AWS, Azure)**, improving DRAGEN multi-omic workflow reliability and supporting **50+** global biopharma customers.
- Led development of **Django-based full-stack applications** for **data visualization**, enabling internal teams to troubleshoot customer issues more efficiently, contributing to an estimated **30% reduction in resolution time**.
- Implemented **3 new features** and performed over **5 code reviews** on the DRAGEN **C++ codebase**, optimizing **RNA-seq quantification** and **gene fusion detection** workflows.

University of Michigan

Oct 2019 – Feb 2022 | Ann Arbor, MI

Research Assistant

- Developed robust analysis pipelines for bulk **RNA-seq**, **CHIP-seq**, **ATAC-seq**, **RRBS** and **single-cell** datasets to study epigenetic regulation and generate actionable insights to support brain and prostate cancer research initiatives.
- Implemented a **DenseNet-CNN** deep learning model using **TensorFlow** for instrument detection in cataract surgery videos, achieving a **0.95 F1 score**.

PayPal

Jan 2019 – June 2019 | Chennai, India

Software Engineer Intern

- Designed and developed a **real-time dashboard** using **React**, **Node.js** with **GraphQL APIs** and **SQL** backend, enabling account managers to monitor large batch payout metrics and communicate insights to customers, **increasing engagement** and **product adoption by 50%**.

EDUCATION

University of Michigan | Ann Arbor, MI

May 2021

Master of Science in Bioinformatics

Birla Institute of Technology and Science (BITS), Pilani | Hyderabad, India

Jul 2019

Bachelor of Engineering in Computer Science